Bounded External Generative Complexity:
Predictive Quotients, No-Smuggling, and Minimal Hierarchical Phases

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Abstract

We study a candidate structural restriction on foundational response theories: externally relevant relations should be generated from controlled-complexity exposed states through a shared controlled-complexity response law rather than by independently specified pairwise couplings. We call this hypothesis *bounded external generative complexity* (BEGC).

The construction combines established ideas from predictive-state representations, Markov lumpability, spectral coarse graining, algebraic quotients, latent response models, and finite-resolution model-complexity accounting. Its distinctive restriction is simultaneous control of both exposed-state complexity and shared response-law complexity, together with hereditary closure of the action and law families through one nested quotient hierarchy.

We define finite-resolution external generative complexity relative to a fixed regular model class, thereby excluding a custom-decoder loophole in which arbitrary response data could be hidden inside an uncharged parameterization. For n response-bearing objects with exposed states on an r-dimensional manifold and one universal p-parameter C^1 response law, we prove that the realizable directed-response family has Hausdorff dimension at most

$$nr + p.$$

Hence, whenever

$$nr + p < n(n-1),$$

the BEGC response family has positive Hausdorff codimension relative to generic directed response space. In the Euclidean realization this also implies zero ambient Lebesgue measure.

We define the external-response quotient relative to a declared action/probe family and prove its universal minimality property. Parent dynamics descends to this quotient if and only if external-response equivalence is dynamically invariant. Parent composition descends if and only if the same equivalence is a compositional congruence; when parent composition is associative, the induced quotient composition inherits associativity.

Within finite reversible continuous-time Markov laboratories, we define exact hierarchical BEGC phases using persistence-selected nested quotients. A six-state hidden-flip-equivariant family admits the exact spectral decomposition

$$\operatorname{spec}(-Q) = \{0, 3t, 2s+t\} \cup \operatorname{spec}(H_{\mathrm{fast}}),$$

with persistence hierarchy precisely when

$$t < s, \qquad \lambda_{\min}(H_{\mathrm{fast}}) > 2s + t.$$

An exhaustive computer-assisted enumeration reproduces the known benchmark of 113 associative operations on a labeled three-element set and then filters this class by the quotient conditions relevant here, yielding 61/35/20/6/9/2 under successive restrictions.

Finally, we classify an exact four-state hierarchical phase. For one branch,

$$\operatorname{spec}(-Q) = \{0, 4x, x + 3y, 2a + x + y\},$$

and the persistence-selected hierarchy exists exactly when

$$0 < x < y < a.$$

The model realizes exact reductions

$$4 \longrightarrow 3 \longrightarrow 2,$$

admits an explicit associative parent composition compatible with both reductions, and introduces no new free response-law parameter under the second quotient. This gives sharp realizability of the elementary cardinality lower bound for two strict nontrivial reductions in the declared finite laboratory.

These results establish exclusion power, mathematical nonemptiness, finite classification, and a minimal hierarchical phase for BEGC-like response theories. They do not derive BEGC itself, physical time, spatial geometry, spatial dimension, a preferred physical grade, Gaussian coupling, nearest-neighbor intergrade dynamics, quantum mechanics, gravity, or cosmology.

1 Introduction

A generic finite response theory may assign an independent relation to every ordered pair of effective objects. For n objects this gives $n(n-1)$ independent off-diagonal response values before any additional internal structure is introduced. Such a theory can fit an arbitrary finite response pattern, but precisely because the response table is unrestricted, the pattern may simply have been inserted as data.

The central question of this paper is whether one can formulate a strictly smaller response architecture in which externally relevant relations are *generated* rather than independently assigned.

The candidate principle studied here is:

> **Bounded External Generative Complexity (BEGC-0).** At every physically admissible response level, externally relevant relations are generated from robustly controlled-complexity exposed states by a universal regular response law of robustly controlled nomological complexity. Independent pair-indexed response data are not fundamental inputs.

BEGC-0 is a candidate physical postulate. It is not derived in this paper and is not asserted here as an established law of nature.

The principle constrains the generative complexity of the *response structure being explained*. It does not impose a low-complexity bound on all contingent microscopic or intragrade state data. Any datum that changes the target external-response object must, however, be charged somewhere in the external generative complexity, regardless of whether it is called a law parameter, state datum, probe datum, decoder datum, or boundary datum.

This distinction is essential. Restricting only object-state complexity does not solve the arbitrary-response problem. Even a system of one-state objects can be assigned an arbitrary matrix W_{ab}. No-smuggling therefore requires simultaneous control of

exposed-state complexity

and

response-law complexity.

The analysis is deliberately pre-geometric. We do not assume a spatial metric, a graph distance, a manifold dimension, a spacetime signature, or an identification between hierarchy labels and physical dimension.

The paper has four logical layers:

Layer A: Candidate principle: BEGC-0.

Layer B: General finite mathematics: external-response complexity, response quotients, dynamical descent, compositional descent, and no-smuggling.

Layer C: Exact finite laboratory: reversible continuous-time Markov models with fully analyzable nested quotients.

Layer D: Conditional HSMT interpretation: stable persistent quotient levels as prototype hierarchical grades.

The individual mathematical ingredients have substantial precedents. The intended contribution is their integration into one criterion that requires simultaneously:

1. controlled complexity of the externally exposed state;
2. controlled complexity of the shared response-law family;
3. external-response quotients defined relative to inherited action/probe families;
4. persistence selection;
5. dynamical congruence;
6. compositional congruence;
7. hereditary closure of both the response-law family and the admissible action family;
8. an explicit comparison between the resulting generative complexity and the $O(n^2)$ arbitrary-pair null.

The principal finite results are:

1. A regular response family generated by n exposed r-dimensional states and one p-parameter universal law has Hausdorff dimension at most

$$nr + p,$$

compared with ambient dimension $n(n-1)$ for a generic directed pair-response table.

2. External-response equivalence defines a coarsest sufficient quotient relative to a specified admissible probe/action family.
3. Dynamics descends exactly when response equivalence is dynamically invariant.
4. Composition descends exactly when response equivalence is a compositional congruence, and associativity is inherited.
5. A six-state symmetry-restricted family admits an exact spectral classification.

6. Compatible associative composition on three exposed types is constrained but nonunique.

7. The elementary cardinality lower bound for two strict nontrivial quotient reductions is sharply realized by an explicit four-state model satisfying the full declared finite phase conditions.

The numeral 4 in the last result is solely a finite cardinality bound. It has no demonstrated relation to four-dimensional spacetime, effective $3+1$ physics, historical HSMT grade $N=4$, or any physical dimension.

2 Relation to Existing Work

The ingredients used in the present construction have substantial precedents in several mature literatures. The purpose of BEGC is not to replace these frameworks or to claim priority for their individual components, but to impose a particular joint restriction on how predictive reduction, response generation, composition, and hierarchy fit together.

2.1 Markov aggregation and lumpability

Exact aggregation of Markov chains by partitions satisfying equal block-to-block transition rates is classical. Strong lumpability and related exact and approximate aggregation conditions provide the direct mathematical antecedent of the dynamical-congruence condition used in our finite Markov laboratory [1, 2].

Accordingly, the finite BEGC laboratory does not introduce a new notion of Markov lumpability. It uses strong lumpability as the exact dynamical-closure condition within a larger hierarchy that simultaneously requires persistence selection, response sufficiency, compositional closure, and hereditary generative closure.

2.2 Predictive states and observable-relative macrostates

Computational mechanics constructs causal states by grouping histories that induce identical predictive distributions over futures and characterizes the resulting representation as minimal for prediction [3]. Predictive-state representations likewise encode dynamical state through predictions of observable experiments rather than inaccessible hidden variables [4].

Particularly close to the response-relative quotient used here is the observable-relative macrostate construction of Shalizi and Moore [5]. There the relevant partition is constrained by a specified observation set and by predictive Markovian closure.

The canonical response quotient used below should therefore be understood as part of this broader predictive-state tradition. The additional BEGC restriction is not the existence of a minimal predictive quotient by itself. It is the simultaneous control of the generative complexity of the interobject response law together with hereditary action, law, dynamical, and compositional closure through multiple protected resolutions.

2.3 Spectral coarse graining and metastability

Slow spectral modes and almost-invariant sets are standard tools for identifying metastable macrostates in reversible Markov dynamics. Spectral methods such as Perron cluster analysis and PCCA/PCCA+ identify persistent aggregates through dominant or slow eigenstructure [6, 7].

The persistence-selection condition below uses this established idea. The additional requirement is that the cumulative slow sector coincide with an externally sufficient quotient that simultaneously satisfies the other BEGC closure conditions.

Thus the spectral sector alone does not define an accepted BEGC grade.

2.4 Behavioral equivalence and algebraic quotients

Behavioral equivalence, probabilistic bisimulation, and coalgebraic treatments of stochastic systems provide broad frameworks for identifying states that are indistinguishable by specified behavior and for constructing quotient systems [8].

Likewise, quotienting an associative system by a congruence and inheriting associativity on the quotient are standard facts of semigroup theory [9].

The descent results proved below are therefore included primarily to make the BEGC construction self-contained. Their role here is to enforce that one and the same response-defined quotient is compatible with both dynamics and composition.

2.5 Latent response models and shared kernels

Latent-position network models generate pair relations from lower-dimensional object states through a common link function [10]. Random-dot-product graph models provide a particularly transparent low-rank realization of this idea [11].

More generally, exchangeable-array representations and graphon theory show that very broad classes of relational arrays can be represented using latent variables and a common measurable function or kernel [12, 13].

These precedents are especially important for BEGC because they show that a *shared kernel alone is not a no-smuggling principle*. An unrestricted graphon or other shared measurable kernel may itself have arbitrarily rich functional complexity. Replacing pair-indexed coefficients by one common F can therefore merely relocate the relational information into F.

BEGC consequently constrains both:

$$\text{the complexity of the exposed state}$$

and

$$\text{the robust complexity of the shared response-law family.}$$

This law-side restriction is essential to the no-smuggling criterion.

2.6 Information compression and model description

The information-bottleneck framework formalizes compression that retains task-relevant information [14]. Minimum description length explicitly charges model structure and parameters when comparing explanations [15].

The quantity

$$K_{\mathrm{ext}}^{(\varepsilon)}(R \mid \mathcal{M})$$

introduced below should therefore not be interpreted as a new general theory of description length. It is a problem-specific finite-resolution accounting device used to ensure that external response information cannot be hidden in an uncharged state representation, decoder, action family, or response-law family.

2.7 Effective projection

Feshbach and Feshbach–Schur maps are established operator-theoretic methods for constructing effective operators once a retained sector has already been identified [16, 17].

They are strictly downstream in the present framework. The BEGC problem is not how to eliminate a known complementary sector; it is how a physically meaningful retained response quotient could be selected without inserting the target sector by hand.

2.8 Scope of the present contribution

Individually, lumpability, predictive sufficiency, spectral coarse-graining, algebraic quotients, latent response models, shared kernels, and description-length penalties are established ideas.

The present formulation combines them into a single external-generative-complexity criterion in which:

1. exposed states and the shared response law are both complexity-controlled;
2. the response quotient is defined relative to an inherited action/probe family;
3. accepted hierarchy levels simultaneously satisfy persistence, dynamical congruence, compositional congruence, action heredity, and response-law heredity;
4. the resulting response family is explicitly compared with the quadratic-complexity arbitrary-pair null.

The finite classifications later in the paper study consequences of this joint definition. They do not assert priority for its individual ingredients.

3 The Arbitrary-Response Null

Let

$$V_n = \{1, \dots, n\}$$

label n response-bearing objects.

The unrestricted directed pair-response null assigns

$$W_{ab} \in \mathbb{R}, \qquad a \neq b,$$

independently.

Its continuous parameter dimension is

$$D_{\mathrm{null}}(n) = n(n-1).$$

At finite resolution $\varepsilon > 0$, assume for illustration that each coefficient lies in a fixed bounded interval of diameter C. Resolving each independent entry to accuracy ε requires approximately

$$\left(\frac{C}{\varepsilon}\right)^{n(n-1)}$$

distinguishable tables, giving

$$\log_2 N_{\mathrm{null}}(\varepsilon) = \Theta\left(n^2 \log \frac{1}{\varepsilon}\right).$$

This quadratic response complexity is the null class BEGC is designed to exclude.

3.1 Bounded state complexity alone is insufficient

Suppose every object has only one microscopic state:

$$X_a = \{*\}.$$

The exposed-state complexity is then minimal.

Nevertheless, define

$$R(a, b) = W_{ab}$$

for an arbitrary matrix W.

The model has zero nontrivial exposed-state dimension but still contains $O(n^2)$ independent response coefficients.

Similarly, let

$$z_a \in \{0, 1\}$$

and define

$$R_{ab}(z_a) = W_{ab} z_a.$$

The exposed state contains only one bit, while

$$\{W_{ab}\}$$

still carries $n(n-1)$ freely adjustable pairwise coefficients.

Thus

$$\boxed{\text{bounded exposed-state complexity} \not\Rightarrow \text{bounded response-law complexity.}}$$

BEGC must constrain both.

4 BEGC and Fixed Model Classes

Definition 4.1 (Regular external generative model class)**.** *A regular external generative model class $\mathcal{M}$ contains the structural information that is fixed before fitting a particular target response. It includes:*

1. *an admissible action/probe convention $\mathcal{P}$;*
2. *an exposed-state space $\mathcal{Z}$;*
3. *an admissible universal response-law family*

$$\mathcal{F} = \{F_\theta : \mathcal{Z} \times \mathcal{Z} \to \mathcal{R}\}_{\theta \in \Theta};$$

4. *metrics and operational domains;*
5. *regularity assumptions on the state and law maps;*
6. *decoding conventions;*
7. *any primitive compositional structure not itself derived;*
8. *any other structural datum that can change the externally relevant response object.*

For n exposed objects, a BEGC realization has states

$$z_1, \dots, z_n \in \mathcal{Z}$$

and a single shared law

$$F_\theta \in \mathcal{F}.$$

The pair response is

$$W_{ab} = F_\theta(z_a, z_b).$$

The law parameter θ is not indexed by the bare object pair (a, b).

4.1 Pair-label blindness

Relabeling covariance does not by itself enforce generative economy.

An arbitrary matrix can transform under a permutation P as

$$W \longmapsto PWP^{-1},$$

while retaining $O(n^2)$ independent entries.

The stronger BEGC requirement is that response be determined by physical exposed states and universal law parameters rather than an independently programmable pair label.

5 Finite-Resolution External Generative Complexity

A naive minimization over arbitrary parameterizations creates a custom-decoder loophole. One could define a bespoke map F_W whose definition already contains a target table W, while reporting no adjustable law parameters.

The model class must therefore be fixed or itself charged.

Definition 5.1 (Finite-resolution external generative complexity)**.** *Fix:*

- *a regular model class $\mathcal{M}$;*
- *a response metric $d_{\mathcal{R}}$;*
- *an operational tolerance $\varepsilon > 0$.*

For a target external-response object R, define

$$K_{\mathrm{ext}}^{(\varepsilon)}(R \mid \mathcal{M})$$

to be the minimum number of bits required, within $\mathcal{M}$, to specify

- *the exposed states;*
- *the admissible response-law parameters;*
- *all other response-relevant data charged by $\mathcal{M}$,*

such that the reconstructed response $\widehat{R}$ satisfies

$$d_{\mathcal{R}}(R, \widehat{R}) \le \varepsilon.$$

Throughout this paper all logarithms in complexity expressions are base 2.

If optimization over model classes is permitted, define

$$K_{\mathrm{ext}}^{(\varepsilon)}(R) = \inf_{\mathcal{M}} \left[L(\mathcal{M}) + K_{\mathrm{ext}}^{(\varepsilon)}(R \mid \mathcal{M}) \right],$$

where $L(\mathcal{M})$ is the description cost of the model class.

For all finite theorems below, $\mathcal{M}$ is fixed in advance.

5.1 Covering-number estimate

Let

$$N_{\mathcal{Z}}(\delta)$$

be the covering number of the exposed-state operational domain at resolution δ, and let

$$N_{\mathcal{F}}(\delta)$$

be the corresponding covering number of the admissible response-law family.

Assume the generative response map is uniformly Lipschitz on the declared operational domain.

Then there are constants $c_Z, c_F > 0$, determined by the fixed model class, such that response accuracy ε can be achieved using state/law covering radii proportional to ε. Hence

$$K_{\mathrm{ext}}^{(\varepsilon)}(R \mid \mathcal{M}) \le n \log_2 N_{\mathcal{Z}}(c_Z \varepsilon) + \log_2 N_{\mathcal{F}}(c_F \varepsilon) + O_{\mathcal{M}}(1).$$

In a regular finite-dimensional case with exposed dimension r and law-parameter dimension p,

$$\log_2 N_{\mathcal{Z}}(\varepsilon) = O\left(r \log \frac{1}{\varepsilon} \right),$$

and

$$\log_2 N_{\mathcal{F}}(\varepsilon) = O\left(p \log \frac{1}{\varepsilon} \right),$$

so

$$K_{\mathrm{ext}}^{(\varepsilon)} = O\left[(nr + p) \log \frac{1}{\varepsilon} \right].$$

If the operational ranges grow at most polynomially with n, an additional

$$O((nr + p) \log n)$$

term may appear, still remaining subquadratic for fixed r, p.

6 No-Smuggling Theorems

Theorem 6.1 (Regular BEGC image-dimension bound)**.** *Let $\mathcal{Z}$ be an r-dimensional second-countable C^1 manifold of exposed states and Θ a p-dimensional second-countable C^1 manifold of universal response-law parameters.*

For n objects define

$$\Phi_n : \mathcal{Z}^n \times \Theta \longrightarrow \mathbb{R}^{n(n-1)}$$

by

$$[\Phi_n(z_1, \ldots, z_n, \theta)]_{ab} = F_\theta(z_a, z_b), \qquad a \neq b.$$

Assume Φ_n is C^1. Then

$$\dim_H [\Phi_n(\mathcal{Z}^n \times \Theta)] \leq nr + p.$$

Consequently, if

$$nr + p < n(n-1),$$

the BEGC response family has strictly positive Hausdorff codimension inside the generic directed-response space

$$\mathbb{R}^{n(n-1)}.$$

Proof. The product

$$\mathcal{Z}^n \times \Theta$$

is a second-countable C^1 manifold of dimension

$$nr + p.$$

A C^1 map is locally Lipschitz in coordinate charts. By second-countability, the domain admits a countable atlas. Restrict each chart to an exhaustion by compact subsets. On each such compact restriction, Φ_n is Lipschitz.

Lipschitz maps do not increase Hausdorff dimension. Therefore every compact chart image has Hausdorff dimension at most $nr + p$. Hausdorff dimension of a countable union is the supremum of the dimensions of the members. Hence

$$\dim_H(\operatorname{Im} \Phi_n) \leq nr + p.$$

If

$$nr + p < n(n-1),$$

the image has positive Hausdorff codimension in the ambient $n(n-1)$-dimensional Euclidean response space.

□

Corollary 6.2 (Ambient Lebesgue-null realization)**.** *Under the hypotheses of the preceding theorem, if*

$$nr + p < n(n-1),$$

then the BEGC response family has zero $n(n-1)$-dimensional Lebesgue measure in

$$\mathbb{R}^{n(n-1)}.$$

Proof. Any subset of $\mathbb{R}^N$ whose Hausdorff dimension is strictly smaller than N has zero N-dimensional Lebesgue measure. □

Corollary 6.3 (Bilinear rank bound)**.** *If*

$$W_{ab} = z_a^{\mathsf{T}} K z_b, \qquad z_a \in \mathbb{R}^r,$$

then

$$\operatorname{rank}(W) \leq r.$$

Proof. Let Z be the matrix whose a-th row is z_a^{T}. Then

$$W = ZKZ^{\mathsf{T}}.$$

Hence

$$\operatorname{rank}(W) \le \operatorname{rank}(Z) \le r.$$

□

For a directed bilinear realization

$$W = UKV^{\mathsf{T}},$$

with U, V each having r columns, the same conclusion holds:

$$\operatorname{rank}(W) \le r.$$

Proposition 6.4 (Finite-resolution entropy separation)**.** *Suppose the fixed model class obeys*

$$\log_2 N_{\mathcal{Z}}(\varepsilon) = O\left(r \log \frac{1}{\varepsilon}\right)$$

and

$$\log_2 N_{\mathcal{F}}(\varepsilon) = O\left(p \log \frac{1}{\varepsilon}\right),$$

with constants independent of n, and suppose the response map is uniformly Lipschitz on its operational domain.

Then

$$K_{\text{ext}}^{(\varepsilon)} = O\left[(nr + p) \log \frac{1}{\varepsilon}\right].$$

A generic independently resolved directed table has

$$K_{\text{ext}}^{(\varepsilon)} = \Theta\left[n^2 \log \frac{1}{\varepsilon}\right].$$

The proposition is a finite-resolution expression of the same no-smuggling distinction as the Hausdorff-dimension theorem.

7 Canonical External-Response Quotients

Let I_k denote the state space at hierarchical resolution k. The parent level is

$$I_0 := X.$$

Let $\mathcal{P}_k$ be the physically admissible action/probe family at level k.

For $x \in I_k$, let

$$R_{\mathcal{P}_k}(x)$$

denote its complete response signature to $\mathcal{P}_k$.

Definition 7.1 (External-response equivalence)**.** *Define*

$$x \sim_{\mathcal{P}_k} y$$

if and only if

$$R_{\mathcal{P}_k}(x) = R_{\mathcal{P}_k}(y).$$

The external-response quotient at resolution k is

$$I_{k+1} = I_k / \sim_{\mathcal{P}_k} .$$

Write

$$q_k : I_k \to I_{k+1}$$

for the quotient map.

The word *canonical* is always relative to the declared action family $\mathcal{P}_k$. This observable-relative construction has close precedents in predictive-state and macrostate theory [3, 4, 5]; our use of it is part of the integrated BEGC architecture rather than a claim of novelty for predictive quotienting itself.

Definition 7.2 (Externally sufficient representation)**.** *A map*

$$Z : I_k \to \mathcal{Z}$$

is externally sufficient relative to $\mathcal{P}_k$ if

$$Z(x) = Z(y) \quad \Longrightarrow \quad R_{\mathcal{P}_k}(x) = R_{\mathcal{P}_k}(y).$$

Theorem 7.3 (Universal minimality of the response quotient)**.** *The quotient*

$$I_{k+1} = I_k / \sim_{\mathcal{P}_k}$$

is the coarsest externally sufficient representation.

For every externally sufficient map

$$Z : I_k \to \mathcal{Z},$$

there exists a unique map

$$h : Z(I_k) \to I_{k+1}$$

such that

$$q_k = h \circ Z.$$

Proof. If

$$Z(x) = Z(y),$$

external sufficiency implies

$$R_{\mathcal{P}_k}(x) = R_{\mathcal{P}_k}(y),$$

hence

$$q_k(x) = q_k(y).$$

Therefore q_k is constant on every fiber of Z. Define

$$h(Z(x)) := q_k(x).$$

This is well-defined because equal Z-values imply equal quotient classes.

Uniqueness follows because every point of $Z(I_k)$ is of the form $Z(x)$.

□

Corollary 7.4 (Minimal sufficient coordinates)**.** *A sufficient representation Z is equivalent to the response quotient up to bijection if and only if*

$$R_{\mathcal{P}_k}(x) = R_{\mathcal{P}_k}(y) \quad \Longleftrightarrow \quad Z(x) = Z(y).$$

7.1 Operational quotient metric

If external response distributions carry a statistical distance d_{out}, an operational metric or pseudo-metric on the quotient may be defined by

$$d_k([x],[y]) = \sup_{p \in \mathcal{P}_k} d_{\mathrm{out}}\big(R_p(x), R_p(y)\big).$$

This gives a coordinate-free route to finite-resolution covering numbers of the exposed quotient.

8 Dynamical and Compositional Descent

8.1 Dynamical descent

Let

$$T_s : I_k \to I_k$$

be a parent evolution map or semigroup.

Theorem 8.1 (Dynamical descent)**.** *There exists a well-defined induced evolution*

$$\overline{T}_s : I_{k+1} \to I_{k+1}$$

satisfying

$$q_k \circ T_s = \overline{T}_s \circ q_k$$

if and only if

$$x \sim_{\mathcal{P}_k} y \quad \Longrightarrow \quad T_s x \sim_{\mathcal{P}_k} T_s y.$$

Proof. If $\overline{T}_s$ exists and

$$q_k(x) = q_k(y),$$

then

$$q_k(T_s x) = \overline{T}_s(q_k(x)) = \overline{T}_s(q_k(y)) = q_k(T_s y).$$

Conversely, if the equivalence is preserved, define

$$\overline{T}_s([x]) := [T_s x].$$

The invariance condition guarantees independence from the chosen representative.

□

In the finite Markov laboratory this condition becomes classical strong lumpability; the theorem is stated abstractly so that the same quotient language can be used outside that laboratory.

8.2 Compositional descent

Assume I_k carries an associative parent composition

$$\otimes : I_k \times I_k \to I_k.$$

Theorem 8.2 (Compositional descent and associativity inheritance)**.** *A well-defined induced composition*

$$\mu_k : I_{k+1} \times I_{k+1} \to I_{k+1}$$

satisfying

$$q_k(x \otimes y) = \mu_k(q_k(x), q_k(y))$$

exists if and only if $\sim_{\mathcal{P}_k}$ is a congruence of $\otimes$:

$$x \sim_{\mathcal{P}_k} x', \qquad y \sim_{\mathcal{P}_k} y'$$

implies

$$x \otimes y \sim_{\mathcal{P}_k} x' \otimes y'.$$

If $\otimes$ is associative, then μ_k is associative.

Proof. If μ_k exists, equivalent representatives must produce the same quotient value, so congruence is necessary.

Conversely, define

$$\mu_k([x], [y]) = [x \otimes y].$$

The congruence condition makes this well-defined.

Associativity follows from

$$\begin{aligned} \mu_k(\mu_k([x], [y]), [z]) &= [(x \otimes y) \otimes z] \\ &= [x \otimes (y \otimes z)] \\ &= \mu_k([x], \mu_k([y], [z])). \end{aligned}$$

□

The algebraic fact itself is standard semigroup quotient theory [9]. Its role here is to require that the same physically defined response quotient that closes dynamically also close under the declared parent composition.

9 Resolution-Indexed Action Families

Successively coarser quotients cannot generally be canonical relative to one unchanged complete microscopic probe family. Canonicality must be resolution-indexed.

Let

$$I_0 \xrightarrow{q_0} I_1 \xrightarrow{q_1} \cdots \xrightarrow{q_{d-1}} I_d$$

be an accepted quotient tower.

At level k, the admissible action family is $\mathcal{P}_k$.

The heredity requirement is

$$q_k^* \mathcal{P}_{k+1} \subseteq \mathcal{P}_k,$$

where

$$q_k^* \phi = \phi \circ q_k.$$

Thus every admissible action at the next level pulls back to an admissible action at the current level and depends only on the accepted quotient class.

When action types permit, a maximal induced family can be written

$$\mathcal{P}_{k+1} = \{\phi : \phi \circ q_k \in \mathcal{P}_k\}.$$

No independent probe database may be introduced at each level unless its complexity is charged in

$$K_{\text{ext}}.$$

10 Exact Hierarchical BEGC Phases

We now formalize the finite exact hierarchy used in the Markov laboratory.

Definition 10.1 (Hereditary response-law closure)**.** *Suppose the effective response law at level k is*

$$F_{\theta_k}.$$

The hierarchy has hereditary response-law closure if the next law remains inside the same fixed controlled family, or inside a declared low-dimensional running family, with parameter update

$$\theta_{k+1} = \mathcal{R}_k(\theta_k, \xi_k),$$

where ξ_k contains only admissible resolved data whose complexity is already counted.

Introducing a new unrelated arbitrary response law at each level is not hereditary closure.

10.1 Persistence selection

Let Q be a finite reversible continuous-time Markov generator with positive stationary distribution π. Set

$$A = -Q.$$

Then A is self-adjoint and positive semidefinite in

$$L^2(\pi).$$

Order its eigenvalues as

$$0 = \lambda_0 \le \lambda_1 \le \cdots \le \lambda_{n-1}.$$

Let Π be a strongly lumpable partition into m blocks, and let

$$E_\Pi$$

be the conditional-expectation operator in $L^2(\pi)$ onto functions constant on blocks of Π.

Definition 10.2 (Exact persistence-selected quotient)**.** *The partition Π is an exact persistence-selected rank-m quotient if*

$$\operatorname{Ran}(E_\Pi) = \operatorname{span}\{\varphi_0, \ldots, \varphi_{m-1}\},$$

where φ_j are eigenvectors belonging to the m smallest eigenvalues, and there is a strict protecting gap

$$\lambda_{m-1} < \lambda_m.$$

This condition is closely related to established spectral methods for metastability and coarse graining [6, 7]. The BEGC requirement is stronger only in the sense that the selected slow sector must coincide with the inherited externally sufficient quotient and satisfy the additional closure conditions.

At a degenerate spectral boundary

$$\lambda_{m-1} = \lambda_m,$$

the cumulative slow subspace is not uniquely selected by eigenvalue ordering alone. The exact examples in this paper avoid this degeneracy.

Definition 10.3 (Exact finite hierarchical BEGC phase)**.** *A finite reversible BEGC model possesses an exact hierarchical phase of depth d if there exists a strict quotient tower*

$$I_0 \xrightarrow{q_0} I_1 \xrightarrow{q_1} \cdots \xrightarrow{q_{d-1}} I_d$$

such that, at every level k:

1. *q_k is the external-response quotient relative to the inherited action family $\mathcal{P}_k$;*
2. *q_k is a dynamical congruence;*
3. *q_k is a compositional congruence whenever composition is part of the declared model;*
4. *the quotient is persistence-selected;*
5. *the induced response law satisfies hereditary generative closure;*
6. *the next action family satisfies action heredity.*

The distinctive object in the present work is this *joint* hierarchical criterion rather than any one of the six conditions in isolation.

Perturbative stability is treated separately as a neighborhood property of the exact phase.

11 Reversible Markov Laboratory

Throughout the finite Markov laboratory, Q is written in the row-sum-zero convention, with Q_{ij} the transition rate from state i to state j for $i \neq j$.

Thus

$$Q_{ij} \geq 0 \qquad (i \neq j),$$

and

$$\sum_j Q_{ij} = 0.$$

For a partition

$$\Pi = \{B_1, \ldots, B_m\},$$

strong lumpability requires that for every source block B_a, every target block B_b, and all

$$i, i' \in B_a,$$

one has

$$\sum_{j \in B_b} Q_{ij} = \sum_{j \in B_b} Q_{i'j}.$$

This is the classical finite Markov strong-lumpability condition [1, 2]. It is precisely the condition that resolved transition rates depend only on the resolved state.

For reversible Q, the block-constant invariant subspace is reducing in $L^2(\pi)$, allowing exact comparison with cumulative slow spectral subspaces.

12 Six-State Symmetry-Restricted Classification

Consider

$$I_0 = \{A, B, C\} \times \mathbb{Z}_2$$

with microscopic states

$$A_0, A_1, B_0, B_1, C_0, C_1.$$

Let the hidden-bit flip rate in type $i \in \{A, B, C\}$ be

$$a_i \geq 0.$$

For distinct types i, j, let the cross block be

$$M_{ij} = \begin{pmatrix} u_{ij} & v_{ij} \\ v_{ij} & u_{ij} \end{pmatrix}, \qquad u_{ij}, v_{ij} \geq 0.$$

Define

$$s_{ij} = u_{ij} + v_{ij},$$

and

$$d_{ij} = u_{ij} - v_{ij}.$$

Then

$$|d_{ij}| \leq s_{ij}.$$

Hidden-bit flip symmetry splits the six-dimensional state space into a three-dimensional even sector and a three-dimensional odd sector.

The partition

$$\{A_0, A_1\}, \quad \{B_0, B_1\}, \quad \{C_0, C_1\}$$

is automatically strongly lumpable.

The exposed quotient generator is

$$Q_{\text{type}} = \begin{pmatrix} -(s_{AB} + s_{AC}) & s_{AB} & s_{AC} \\ s_{AB} & -(s_{AB} + s_{BC}) & s_{BC} \\ s_{AC} & s_{BC} & -(s_{AC} + s_{BC}) \end{pmatrix}.$$

For the further quotient

$$\{A, B\} \mid \{C\}$$

to be strongly lumpable, one must have

$$s_{AC} = s_{BC}.$$

Write

$$s_{AB} = s, \qquad s_{AC} = s_{BC} = t.$$

Then

$$Q_3 = \begin{pmatrix} -(s+t) & s & t \\ s & -(s+t) & t \\ t & t & -2t \end{pmatrix}.$$

Its exposed eigenmodes are

$$(1,1,1), \qquad (1,1,-2), \qquad (1,-1,0),$$

with eigenvalues of $-Q_3$

$$0, \qquad 3t, \qquad 2s+t.$$

The odd hidden-parity block of $-Q$ is

$$H_{\text{fast}} = \begin{pmatrix} 2a_A + s + t & -d_{AB} & -d_{AC} \\ -d_{AB} & 2a_B + s + t & -d_{BC} \\ -d_{AC} & -d_{BC} & 2a_C + 2t \end{pmatrix}.$$

Theorem 12.1 (Six-state spectral classification)**.** *For the reversible hidden-flip-equivariant family above,*

$$\operatorname{spec}(-Q) = \{0, 3t, 2s+t\} \cup \operatorname{spec}(H_{\text{fast}}).$$

The exact quotient tower

$$6 \longrightarrow 3 \longrightarrow 2$$

is persistence ordered precisely when

$$t < s$$

and

$$\lambda_{\min}(H_{\text{fast}}) > 2s + t.$$

The two protection gaps are

$$\Delta_{\text{slow}} = 2(s-t),$$

and

$$\Delta_{\text{fast}} = \lambda_{\min}(H_{\text{fast}}) - (2s+t).$$

Proof. The hidden-bit flip operator commutes with Q. Therefore the six-dimensional representation decomposes into even and odd invariant subspaces.

The even block is precisely the three-state quotient Q_3, with eigenvalues

$$0,\ 3t,\ 2s+t.$$

The odd block is

$$-H_{\text{fast}}.$$

Therefore the full spectrum is the union of the two sector spectra.

The second quotient retains the slowest nonstationary exposed mode precisely when

$$3t < 2s + t,$$

equivalent to

$$t < s.$$

All three exposed modes lie below every hidden mode precisely when

$$2s + t < \lambda_{\min}(H_{\text{fast}}).$$

The gap formulas follow by subtraction.

□

This theorem is an analytic classification of the specific symmetry-restricted family above, not a new general theorem of metastability.

Full numerical matrices are given in Appendix A.

13 Three-Type Associative Composition Enumeration

Let

$$S = \{A, B, C\}.$$

There are

$$3^9 = 19683$$

labeled binary operations

$$\mu : S \times S \to S.$$

The known number of associative binary operations on a labeled three-element set is

$$113,$$

equivalently the number of labeled semigroups of order three [18]. Our enumeration first reproduces this known benchmark and then imposes the quotient predicates relevant to the present hierarchy.

Fix the quotient relation

$$A \sim B, \qquad C \text{ separate.}$$

Proposition 13.1 (Computational exhaustive quotient filtering)**.** *Exhaustive enumeration of all* 19683 *labeled binary operations on* S *reproduces the known* 113 *associative operations.*

Among these,

$$61$$

respect

$$A \sim B \mid C$$

as a congruence.

Among those 61*:*

$$35$$

are commutative,

$$20$$

are idempotent,

$$6$$

are both commutative and idempotent,

$$9$$

have absorbing C*,*

and

$$2$$

are simultaneously quotient-compatible, commutative, idempotent, and have absorbing C*.*

The final two multiplication tables are

	A	B	C
A	A	A	C
B	A	B	C
C	C	C	C

and

	A	B	C
A	A	B	C
B	B	B	C
C	C	C	C.

They are the two chain-semilattice orientations under interchange of the relative order of A and B, with C absorbing.

The proposition establishes two facts:

1. hierarchical congruence genuinely restricts the associative-law space;
2. quotient hierarchy plus associativity does not uniquely determine the parent composition law.

The total 113 is therefore not presented as a new enumeration result. The paper-specific computation is the filtering of that known associative class by the declared quotient conditions.

The complete deterministic enumeration script is reproduced in Appendix B.

14 Minimal Four-State Hierarchical Phase

We now consider

$$I_0 = \{A_0, A_1, B, C\}.$$

Let the reversible symmetric transition rates be

$$A_0 \leftrightarrow A_1 : a,$$

$$A_h \leftrightarrow B : x, \qquad h = 0, 1,$$

$$A_h \leftrightarrow C : y, \qquad h = 0, 1,$$

and

$$B \leftrightarrow C : z.$$

Thus

$$Q = \begin{pmatrix} -(a+x+y) & a & x & y \\ a & -(a+x+y) & x & y \\ x & x & -(2x+z) & z \\ y & y & z & -(2y+z) \end{pmatrix}.$$

The first quotient

$$\{A_0, A_1\} \mid \{B\} \mid \{C\}$$

is automatically strongly lumpable, yielding

$$Q_3 = \begin{pmatrix} -(x+y) & x & y \\ 2x & -(2x+z) & z \\ 2y & z & -(2y+z) \end{pmatrix}.$$

14.1 Branch I: merge A with C

Require the second quotient

$$\{A, C\} \mid \{B\}.$$

Strong lumpability requires equal rates from A and C into B:

$$z = x.$$

Therefore

$$Q_3 = \begin{pmatrix} -(x+y) & x & y \\ 2x & -3x & x \\ 2y & x & -(2y+x) \end{pmatrix}.$$

Theorem 14.1 (Four-state spectral classification)**.** *For*

$$z = x,$$

the full spectrum is

$$\operatorname{spec}(-Q) = \{0, 4x, x+3y, 2a+x+y\}.$$

The quotient tower

$$4 \longrightarrow 3 \longrightarrow 2$$

is strictly cumulative-slow and persistence selected precisely when

$$0 < x < y < a.$$

The protection gaps are

$$\Delta_{\text{slow}} = 3(y-x)$$

and

$$\Delta_{\text{fast}} = 2(a-y).$$

Proof. The hidden vector

$$v_h = (1, -1, 0, 0)^{\mathsf{T}}$$

satisfies

$$-Qv_h = (2a+x+y)v_h.$$

The remaining three-dimensional block-constant sector is governed by Q_3. For $z = x$, direct evaluation gives

$$\det(\lambda I + Q_3) = \lambda(\lambda - 4x)(\lambda - x - 3y).$$

Hence

$$\operatorname{spec}(-Q) = \{0, 4x, x+3y, 2a+x+y\}.$$

Strict cumulative-slow ordering requires

$$4x < x+3y < 2a+x+y.$$

The first inequality is

$$x < y.$$

The second is

$$y < a.$$

Together with $x > 0$,

$$0 < x < y < a.$$

Finally,

$$(x + 3y) - 4x = 3(y - x),$$

and

$$(2a + x + y) - (x + 3y) = 2(a - y).$$

□

The exact second quotient is

$$Q_2 = \begin{pmatrix} -x & x \\ 3x & -3x \end{pmatrix}.$$

Its unique nonzero eigenvalue is

$$4x.$$

Thus the slowest nonstationary eigenvalue is inherited exactly through

$$4 \longrightarrow 3 \longrightarrow 2.$$

14.2 Stationary weights

Because the microscopic four-state generator is symmetric, the stationary microscopic measure is uniform.

At the first quotient,

$$\pi_A = \frac{1}{2}, \qquad \pi_B = \frac{1}{4}, \qquad \pi_C = \frac{1}{4}.$$

At the second quotient

$$D = \{A, C\},$$

so

$$\pi_D = \frac{3}{4}, \qquad \pi_B = \frac{1}{4}.$$

Detailed balance for Q_2 is

$$\frac{3}{4}x = \frac{1}{4}(3x).$$

14.3 Concrete numerical witness

Choose

$$a = 10, \qquad x = 0.05, \qquad y = 0.10, \qquad z = 0.05.$$

Then

$$Q = \begin{pmatrix} -10.15 & 10 & 0.05 & 0.10 \\ 10 & -10.15 & 0.05 & 0.10 \\ 0.05 & 0.05 & -0.15 & 0.05 \\ 0.10 & 0.10 & 0.05 & -0.25 \end{pmatrix}.$$

Its spectrum is

$$\operatorname{spec}(-Q) = \{0, 0.20, 0.35, 20.15\}.$$

The gaps are

$$\Delta_{\mathrm{slow}} = 0.15$$

and

$$\Delta_{\mathrm{fast}} = 19.80.$$

The exact three-state quotient is

$$Q_3 = \begin{pmatrix} -0.15 & 0.05 & 0.10 \\ 0.10 & -0.15 & 0.05 \\ 0.20 & 0.05 & -0.25 \end{pmatrix},$$

with

$$\mathrm{spec}(-Q_3) = \{0, 0.20, 0.35\}.$$

The exact two-state quotient is

$$Q_2 = \begin{pmatrix} -0.05 & 0.05 \\ 0.15 & -0.15 \end{pmatrix},$$

with

$$\mathrm{spec}(-Q_2) = \{0, 0.20\}.$$

14.4 Other branches

The other possible second mergers give analogous persistence regions.

For

$$\{A, B\} \mid \{C\},$$

strong lumpability requires

$$z = y,$$

and persistence selection holds for

$$0 < y < x < a.$$

For

$$\{B, C\} \mid \{A\},$$

strong lumpability requires

$$x = y =: u,$$

and persistence selection holds for

$$0 < u < z < a.$$

Thus the four-state phase is not restricted to one quotient topology.

15 Associative Parent Composition for the Four-State Phase

Order the exposed types as

$$A < C < B.$$

Define quotient composition by

$$\mu(s,t) = \max(s,t).$$

This is associative, commutative, and idempotent.

The second quotient

$$A \sim C, \qquad B \text{ separate},$$

is a congruence of μ.

We now lift the operation to all four microscopic states.

Define

$\star$	A_0	A_1	C	B
A_0	A_0	A_1	C	B
A_1	A_1	A_0	C	B
C	C	C	C	B
B	B	B	B	B

.

Proposition 15.1 (Associative microscopic parent product)**.** *The operation $\star$ is associative and commutative.*

Proof. Commutativity is immediate from the table.

For associativity, consider three factors.

If at least one factor is B, every parenthesization yields B.

If no factor is B but at least one is C, every parenthesization yields C.

If all three factors lie in

$$\{A_0, A_1\},$$

the operation is XOR on the hidden $\mathbb{Z}_2$ bit, which is associative.

These cases exhaust all triples.

□

Define

$$q_0(A_0) = q_0(A_1) = A, \qquad q_0(C) = C, \qquad q_0(B) = B.$$

Then

$$q_0(x \star y) = \max(q_0(x), q_0(y)).$$

Define the second quotient

$$q_1(A) = q_1(C) = D, \qquad q_1(B) = E.$$

The induced two-state operation is

$$D \star D = D,$$
$$D \star E = E,$$
$$E \star D = E,$$
$$E \star E = E.$$

Thus both quotient relations are compositional congruences.

The existence of such a product is part of the finite realizability construction. It is not claimed that BEGC uniquely selects this composition; the three-type enumeration above shows explicitly that composition remains nonunique under the quotient constraints alone.

16 Hereditary Law Closure

For the branch

$$z = x,$$

the exposed three-state generator is controlled by the two parameters

$$(x, y).$$

After the second quotient, the effective two-state law depends only on

$$x.$$

Therefore the response-law parameter update is

$$(x, y) \longmapsto x.$$

No new free response-law parameter appears.
This is exact hereditary generative closure in the finite witness.

17 Sharp Realizability of the Four-State Cardinality Bound

The lower-bound part of the following result is elementary. The paper-specific content is that the bound can be saturated while simultaneously satisfying all of the declared finite phase conditions.

Theorem 17.1 (Sharp realizability of the two-grade cardinality bound)**.** *Suppose an exact finite hierarchical phase contains two strict nontrivial quotient maps*

$$I_0 \longrightarrow I_1 \longrightarrow I_2$$

and the final retained quotient satisfies

$$|I_2| \geq 2.$$

Then

$$|I_0| \geq 4.$$

The explicit reversible construction above realizes

$$4 \longrightarrow 3 \longrightarrow 2$$

while also satisfying persistence selection, dynamical congruence, compatible associative parent composition, compositional congruence, and hereditary response-law closure.

Therefore the elementary cardinality lower bound is sharp in the declared exact finite BEGC laboratory.

Proof. Strict reduction implies

$$|I_0| > |I_1| > |I_2|.$$

Because

$$|I_2| \geq 2,$$

we must have

$$|I_1| \geq 3$$

and therefore

$$|I_0| \geq 4.$$

The explicit four-state phase constructed above saturates the bound and supplies the additional BEGC phase properties stated in the theorem.

□

Remark 17.2 (Firewall on the integer 4)**.** *The result*

$$n_{\min} = 4$$

is solely a state-cardinality statement for the declared finite quotient problem. It has no demonstrated relation to spatial dimension, $3+1$ *spacetime, historical HSMT grade* $N = 4$*, or any physical grade number.*

18 Perturbative Defects and Approximate Phases

Let

$$L_0$$

be a finite self-adjoint reference generator and

$$P$$

an exact reducing spectral projector:

$$[L_0, P] = 0.$$

Define

$$P^{\perp} = I - P$$

and perturb

$$L = L_0 + E, \qquad \|E\| \leq \varepsilon.$$

Three distinct perturbative quantities must be separated.

18.1 Spectral-subspace rotation

Suppose the retained spectral cluster is separated from the discarded spectrum by

$$\Delta > 0.$$

For sufficiently small ε/Δ, the perturbed spectral projector $\widetilde{P}$ satisfies

$$\|\widetilde{P} - P\| = O\Big(\frac{\varepsilon}{\Delta}\Big).$$

One way to obtain the scaling is through the Riesz projector

$$P = \frac{1}{2\pi i} \oint_{\Gamma} (z - L_0)^{-1}\, dz,$$

where Γ encloses the retained cluster and remains $O(\Delta)$ away from the spectrum.

Using

$$(z - L)^{-1} - (z - L_0)^{-1} = (z - L)^{-1} E (z - L_0)^{-1},$$

one obtains

$$\|\widetilde{P} - P\| = O(\varepsilon/\Delta).$$

18.2 Direct leakage

Because

$$P^{\perp} L_0 P = 0,$$

we have

$$P^{\perp} L P = P^{\perp} E P.$$

Therefore

$$\|P^{\perp} L P\| \le \varepsilon.$$

Direct leakage is first order in perturbation magnitude:

$$O(\varepsilon).$$

18.3 Feshbach memory/self-energy

For spectral parameter ζ away from the eliminated spectrum, the Feshbach correction is

$$\Sigma(\zeta) = P L P^{\perp} \big(\zeta - P^{\perp} L P^{\perp}\big)^{-1} P^{\perp} L P.$$

If

$$\operatorname{dist}\Big(\zeta, \operatorname{spec}(P^{\perp} L P^{\perp})\Big) \ge \Delta_{\zeta},$$

then

$$\|\Sigma(\zeta)\| \le \frac{\|P L P^{\perp}\| \|P^{\perp} L P\|}{\Delta_{\zeta}} = O\left(\frac{\varepsilon^2}{\Delta_{\zeta}}\right).$$

Thus

$$\boxed{\begin{aligned} \text{spectral-subspace rotation} &= O(\varepsilon/\Delta), \\ \text{direct leakage} &= O(\varepsilon), \\ \text{Feshbach memory/self-energy} &= O(\varepsilon^2/\Delta_{\zeta}). \end{aligned}}$$

The Feshbach construction is an established downstream effective-sector tool [16, 17]. Its use here does not select the physical quotient; it only characterizes the effective consequences once a retained sector has already been identified.

These are distinct defect observables, not three names for one width.

18.4 Congruence defect in the four-state model

Exact second-level lumpability requires

$$z = x.$$

Perturb

$$z = x + \delta.$$

Then

$$A \to B = x,$$

while

$$C \to B = x + \delta.$$

The second-level lumpability defect is therefore directly controlled by

$$|\delta|.$$

This defect is distinct from the spectral protection parameters

$$y - x$$

and

$$a - y.$$

A hierarchy can therefore fail through at least three separate mechanisms:

1. loss of spectral protection;
2. loss of dynamical congruence;
3. loss of compositional congruence.

19 Conditional HSMT Interpretation

The finite mathematics may be interpreted conditionally as an upstream response mechanism for hierarchical effective structure.

A layout-safe conceptual chain is

BEGC parent response $\longrightarrow$ external-response quotients $\longrightarrow$ persistent congruent hierarchy
$\longrightarrow$ grades $\longrightarrow$ linear realization/projectors $\longrightarrow$ projected effective dynamics.

This interpretation is conditional and does not promote BEGC-0 to an established physical law.

19.1 Quotients before projectors

The abstract structure produces

equivalence relations

and

quotient maps.

An orthogonal projector P_N requires an additional linear or Hilbert-space realization.

Thus projectors are representation-dependent realizations of quotient structure, not primitive consequences of the abstract response theory.

19.2 Grade is not dimension

Nothing in BEGC identifies a hierarchy grade with spatial dimension.

If an effective dimension exists, it must be independently characterized by an appropriate scaling observable inside the effective level.

No equality

$$N = d$$

is assumed or derived.

19.3 Shell-like fuzziness is multi-source

Approximate hierarchy may be degraded by

spectral-gap loss,

dynamical-congruence leakage,

compositional-congruence defect,

and

memory from eliminated structure.

No Gaussian width law follows from the present analysis.

19.4 Immanence as factor dynamics

If a finer evolution $T^{(f)}$ and a coarser evolution $T^{(c)}$ satisfy

$$qT^{(f)} = T^{(c)}q,$$

then the coarser system is a dynamical factor of the finer one.

This supplies a precise possible meaning for one mechanics being "immanent" in a richer mechanics without implying geometric containment.

20 Limitations and Claim Status

The present theorem package does not derive BEGC-0.

Several limitations remain essential.

1. **Origin of BEGC.**

   No weaker pre-geometric principle has been shown to force bounded external generative complexity.

2. **Action/probe class.**

   The admissible external action family belongs to the parent model. Successively coarser probe families must be inherited through accepted quotients rather than selected freely.

3. **Predictive quotienting is not claimed as a new idea.**

   Minimal predictive states, predictive-state representations, and observable-relative macrostates have established precedents [3, 4, 5]. The BEGC claim concerns their integration with law-side complexity and hereditary hierarchy.

4. **Composition is not uniquely selected.**

   The order-three enumeration proves that quotient compatibility constrains associative composition but leaves many admissible laws.

5. **Shared kernels are not enough.**

   Latent-variable, exchangeable-array, and graphon models demonstrate that a common kernel may itself contain large functional complexity [10, 12, 13]. BEGC therefore requires law-side complexity control in addition to shared functional form.

6. **Finite reversible laboratory.**

   The exact classifications in this paper do not yet extend automatically to nonreversible, infinite-dimensional, non-Markovian, or quantum systems.

7. **Geometry.**

   No spatial metric, spacetime manifold, locality law, or physical dimension has been derived.

8. **Physical time.**

   The semigroup parameter in the finite Markov laboratory is not identified here with fundamental physical time.

9. **Preferred grade.**

   No physically preferred hierarchy label or grade number is derived.

10. **Gaussian overlap.**

    No Gaussian intergrade kernel, overlap law, or width is derived.

11. **Nearest-neighbor intergrade response.**

    Approximate quotient structure does not force coupling only between adjacent grades.

12. **Downstream physical theories.**

    Quantum theory, gauge structure, gravity, and cosmology are not derived by the present theorem package.

13. **Empirical prediction.**

    The exact spectral identities derived here are mathematical predictions inside the declared finite models, not yet direct observational predictions.

20.1 Claim-status table

Status	Claims
Established within the declared finite mathematics	BEGC image-dimension/no-smuggling theorem; restricted Lebesgue-null corollary; response-quotient universal property; dynamical descent iff congruence; compositional descent iff congruence; associativity inheritance; six-state spectral classification; reproduction of the known 113 order-three semigroup benchmark and quotient-specific filter counts; four-state exact persistence-selected hierarchy; explicit compatible parent composition; sharp realizability of the four-state cardinality bound.
Established antecedents used here	Strong lumpability; predictive sufficient states; spectral metastability and coarse graining; semigroup congruence; latent-state and shared-kernel response models; description-length accounting; Feshbach–Schur effective reduction.
Conditional / model-class dependent	Finite-resolution covering constants; operational action-family hierarchy; particular response-law family; perturbative constants; regularity class used for complexity bounds; interpretation of stable quotients as HSMT grades.
Candidate physical postulate	BEGC-0.
Not derived	Physical time; spatial geometry; spatial dimension; physical or historical $N = 4$; $3 + 1$ spacetime; Gaussian overlap; nearest-neighbor intergrade law; quantum dynamics; gauge structure; gravity; cosmology.

21 Discussion

The main result of this paper is not that hierarchy is automatic, nor that predictive quotienting, lumpability, spectral coarse graining, or latent-state response are individually new.

Rather, the paper studies a joint structural restriction.

First, externally relevant interaction must have bounded generative complexity. Exposed objects are described by controlled-complexity states, and their interaction is generated by one shared controlled-complexity law rather than an independent pairwise table.

Second, the realized system must lie in a hierarchical phase in which successive external-response quotients are persistence-selected and simultaneously compatible with dynamics, composition, action heredity, and response-law heredity.

The finite models show these requirements are mutually compatible.

They also separate several conditions that are easy to conflate.

Low-dimensional exposed state alone is insufficient because arbitrary law data can remain in W_{ab}.

Relabeling covariance alone is insufficient because an arbitrary matrix can transform covariantly.

A shared kernel alone is insufficient because graphon and exchangeability theory demonstrate that a single common measurable function may itself carry very rich relational structure.

Exact lumpability alone is insufficient because a quotient can retain a faster mode while discarding a slower one.

Associativity alone is insufficient because many associative operations remain compatible with the same quotient hierarchy.

A new unrestricted response law at every level defeats recursive generative explanation even if every individual level has a small state space.

The four-state model sharply realizes the elementary minimum cardinality compatible with two strict nontrivial reductions while simultaneously satisfying the declared persistence, dynamical, compositional, and hereditary conditions.

Its derived ordering

$$0 < x < y < a$$

and spectrum

$$0, \quad 4x, \quad x + 3y, \quad 2a + x + y$$

provide a compact example of predictive surplus within the finite model.

The six-state family supplies a complementary result: symmetry can protect an exact quotient architecture while persistence inequalities select its spectral ordering.

The order-three enumeration adds an equally important negative result: hierarchical congruence narrows the composition-law space but does not determine it uniquely.

Thus the finite program establishes

$$\text{exclusion},$$

$$\text{nonemptiness},$$

$$\text{classification},$$

and

$$\text{sharp realizability},$$

but not a complete theory of physical reality.

22 Conclusion

We introduced bounded external generative complexity as a candidate restriction on foundational response theories.

The proposal combines established ideas of predictive compression, Markov aggregation, spectral persistence, quotient closure, latent response, and complexity accounting, but imposes a joint restriction not contained in any one of those ingredients separately:

$$\text{controlled exposed state}$$

together with

$$\text{controlled shared response law},$$

with both the admissible action family and the law family required to descend hereditarily through a common persistence-selected quotient hierarchy.

Relative to a fixed regular model class, this produces a precise no-smuggling theorem. For n objects carrying r-dimensional exposed states and interacting through a p-parameter universal law, the realizable response family has Hausdorff dimension at most

$$nr + p,$$

rather than the

$$n(n-1)$$

dimensions of a generic directed response table.

External-response equivalence gives a minimal quotient relative to a declared action family. Dynamics descends exactly when that equivalence is dynamically invariant, and composition descends exactly when the equivalence is a compositional congruence. Associativity is then inherited rather than separately postulated at the quotient level.

In reversible finite Markov laboratories we classified exact nested persistence-selected quotients, obtained a symmetry-restricted six-state spectral theorem, reproduced the known 113 labeled semigroups of order three before applying quotient-specific filters, and constructed a four-state model that sharply realizes the elementary minimum cardinality for two strict nontrivial nested levels while satisfying the full declared finite phase conditions.

The finite result

$$n_{\min} = 4$$

is purely a cardinality statement and carries no geometric or historical interpretation.

The next research problems are no longer to invent another upstream principle, but to test and generalize the present one:

- extend BEGC beyond reversible finite Markov systems;
- determine which response-law families admit robust hierarchical phases;
- classify larger branching structures and their generative complexities;
- determine whether stronger predictive identities follow from BEGC;
- investigate operator and quantum generalizations without reintroducing pairwise smuggling;
- identify empirical structures, if any, that could distinguish a BEGC-compatible parent theory from a generic response theory.

BEGC-0 remains a candidate physical postulate. The results established here provide a finite mathematical framework in which that candidate can be classified, falsified, and developed without assuming spatial geometry, physical time, target dimension, Gaussian overlap, or nearest-neighbor intergrade dynamics.

A Six-State Detailed Matrices

For the concrete six-state witness, order the states as

$$A_0, A_1, B_0, B_1, C_0, C_1.$$

Take

$$a_A = a_B = a_C = 10,$$
$$s = 0.10, \qquad t = 0.05,$$

and

$$d_{AB} = 0.10, \qquad d_{AC} = d_{BC} = 0.05.$$

$$Q_6 = \begin{pmatrix} -10.15 & 10 & 0.10 & 0 & 0.05 & 0 \\ 10 & -10.15 & 0 & 0.10 & 0 & 0.05 \\ 0.10 & 0 & -10.15 & 10 & 0.05 & 0 \\ 0 & 0.10 & 10 & -10.15 & 0 & 0.05 \\ 0.05 & 0 & 0.05 & 0 & -10.10 & 10 \\ 0 & 0.05 & 0 & 0.05 & 10 & -10.10 \end{pmatrix}.$$

It is symmetric, so the stationary microscopic distribution is uniform.

Its exact spectrum is

$$\operatorname{spec}(-Q_6) = \left\{0, \frac{3}{20}, \frac{1}{4}, 20, \frac{403}{20}, \frac{81}{4}\right\}.$$

Equivalently,

$$\operatorname{spec}(-Q_6) = \{0, 0.15, 0.25, 20, 20.15, 20.25\}.$$

The first quotient is

$$Q_3 = \begin{pmatrix} -\frac{3}{20} & \frac{1}{10} & \frac{1}{20} \\ \frac{1}{10} & -\frac{3}{20} & \frac{1}{20} \\ \frac{1}{20} & \frac{1}{20} & -\frac{1}{10} \end{pmatrix},$$

with

$$\operatorname{spec}(-Q_3) = \left\{0, \frac{3}{20}, \frac{1}{4}\right\}.$$

The second quotient is

$$Q_2 = \begin{pmatrix} -\frac{1}{20} & \frac{1}{20} \\ \frac{1}{10} & -\frac{1}{10} \end{pmatrix},$$

with

$$\operatorname{spec}(-Q_2) = \left\{0, \frac{3}{20}\right\}.$$

Hence

$$\Delta_{\text{slow}} = \frac{1}{4} - \frac{3}{20} = \frac{1}{10},$$

and

$$\Delta_{\text{fast}} = 20 - \frac{1}{4} = \frac{79}{4} = 19.75.$$

B Reproducibility Script for the Order-Three Enumeration

The following deterministic script enumerates all

$$3^9 = 19683$$

binary operations on three labeled elements and evaluates the predicates used in the computational proposition.

It first reproduces the established benchmark

$$113$$

for labeled semigroups of order three [18], providing a useful independent check that the exhaustive enumeration is functioning as intended.

```
from itertools import product

S = range(3) # A=0, B=1, C=2

def op(table, a, b):
    return table[3*a + b]

def associative(table):
    return all(
```

```
        op(table, op(table, a, b), c)
        ==
        op(table, a, op(table, b, c))
        for a in S for b in S for c in S
    )

def commutative(table):
    return all(
        op(table, a, b) == op(table, b, a)
        for a in S for b in S
    )

def idempotent(table):
    return all(
        op(table, a, a) == a
        for a in S
    )

def coarse(x):
    # quotient A~B, C separate
    return 0 if x in (0, 1) else 1

def quotient_congruence(table):
    for a in S:
        for ap in S:
            if coarse(a) != coarse(ap):
                continue
            for b in S:
                for bp in S:
                    if coarse(b) != coarse(bp):
                        continue
                    lhs = coarse(op(table, a, b))
                    rhs = coarse(op(table, ap, bp))
                    if lhs != rhs:
                        return False
    return True

def C_absorbing(table):
    C = 2
    return all(
        op(table, C, x) == C
        and
        op(table, x, C) == C
        for x in S
    )

assoc = []

for table in product(S, repeat=9):
    if associative(table):
        assoc.append(table)

compatible = [
    t for t in assoc
    if quotient_congruence(t)
]

print("associative:", len(assoc))
print("compatible:", len(compatible))
```

```
print(
    "commutative:",
    sum(commutative(t) for t in compatible)
)

print(
    "idempotent:",
    sum(idempotent(t) for t in compatible)
)

print(
    "commutative + idempotent:",
    sum(
        commutative(t) and idempotent(t)
        for t in compatible
    )
)

print(
    "absorbing C:",
    sum(C_absorbing(t) for t in compatible)
)

print(
    "commutative + idempotent + absorbing C:",
    sum(
        commutative(t)
        and idempotent(t)
        and C_absorbing(t)
        for t in compatible
    )
)
```

The output is

```
associative: 113
compatible: 61
commutative: 35
idempotent: 20
commutative + idempotent: 6
absorbing C: 9
commutative + idempotent + absorbing C: 2
```

C Four-State Detailed Calculations

For the branch

$$z = x,$$

the generator is

$$Q = \begin{pmatrix} -(a+x+y) & a & x & y \\ a & -(a+x+y) & x & y \\ x & x & -3x & x \\ y & y & x & -(2y+x) \end{pmatrix}.$$

The hidden vector

$$v_h = (1, -1, 0, 0)^{\mathsf{T}}$$

obeys

$$-Qv_h = (2a + x + y)v_h.$$

A block-constant vector has the form

$$(\alpha, \alpha, \beta, \gamma)^{\mathsf{T}}.$$

The induced three-state generator is

$$Q_3 = \begin{pmatrix} -(x+y) & x & y \\ 2x & -3x & x \\ 2y & x & -(2y+x) \end{pmatrix}.$$

The characteristic polynomial of $-Q_3$ is

$$\det(\lambda I - (-Q_3)) = \det(\lambda I + Q_3) = \lambda(\lambda - 4x)(\lambda - x - 3y).$$

Hence

$$\operatorname{spec}(-Q_3) = \{0, 4x, x + 3y\}.$$

Together with the hidden eigenvalue,

$$\operatorname{spec}(-Q) = \{0, 4x, x + 3y, 2a + x + y\}.$$

The strict ordering

$$0 < 4x < x + 3y < 2a + x + y$$

is equivalent to

$$0 < x < y < a.$$

The two exact protection gaps are

$$\Delta_{\text{slow}} = 3(y - x),$$

and

$$\Delta_{\text{fast}} = 2(a - y).$$

The exact second quotient is

$$Q_2 = \begin{pmatrix} -x & x \\ 3x & -3x \end{pmatrix}.$$

Its unique nonzero eigenvalue is

$$4x.$$

For

$$a = 10, \qquad x = 0.05, \qquad y = 0.10,$$

the exact spectrum is

$$\{0, 0.20, 0.35, 20.15\}.$$

D Model-Class Description Cost

The finite theorems in this paper fix the admissible model class $\mathcal{M}$ in advance.

The notation

$$K_{\mathrm{ext}}^{(\varepsilon)}(R \mid \mathcal{M})$$

therefore does not attempt to define an absolute algorithmic complexity of physical response.

If a future analysis compares multiple model classes, one may use

$$K_{\mathrm{ext}}^{(\varepsilon)}(R) = \inf_{\mathcal{M}} \left[L(\mathcal{M}) + K_{\mathrm{ext}}^{(\varepsilon)}(R \mid \mathcal{M}) \right].$$

The term

$$L(\mathcal{M})$$

is required because otherwise an arbitrary target response could be hidden in a custom decoder or custom response family and then reported as a zero-parameter law.

This is directly analogous in spirit to established description-length accounting [15]. The present quantity is only a response-specific operationalization of the BEGC firewall.

None of the finite dimension or quotient theorems depends on a particular universal coding scheme.

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